INCH-POUND
MIL-R-24183A(SH)
5 August 1988
SUPERSEDING
MIL-R-24183(SHIPS)
29 November 1965
(See 6.6)

MILITARY SPECIFICATION

ROPE AND YARN, PLIED, SYNTHETIC FIBER

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rope and yarn to be used as ratline, seizing, marline, houseline and roundline.

1.2 Classification. Ropes and plied yarns covered by this specification shall be of the following types and classes, as specified (see 6.2):

Type I - Rope.
Class 1 - Ratline and seizing.

Type II - Yarn, plied.
Class 1 - Marline.
Class 2 - Houseline and roundline.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020
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SPECIFICATION

MILITARY
MIL-C-3131 - Cordage; Packaging of.

STANDARDS

FEDERAL
FED-STD-191 - Textile Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The ropes and plied yarns covered by this specification shall be fabricated wholly of virgin synthetic fibers, except nylon and fibers of cellulosic origin. Synthetic fibers of the continuous monofilament or continuous multifilament types may be utilized but not in combination with each other. The size of the monofilament fibers shall be not greater than 200 denier per filament while the size of the multifilament fibers shall be not less than 4.4 denier. In addition, each fiber type shall be fortified with heat and light stabilizers.

3.2 Construction.

3.2.1 Ropes. The ropes, of right hand of Z lay construction, shall consist of three strands, each of which shall be of one size of balanced-twist three-ply yarns and shall conform to the requirements of table I.

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TABLE I. Ratline and seizing, type I.

Circ. (inches)	Tolerance (plus or minus) (inches)	Load "P" 200 X D^2 (pounds)	Feet per pound (min.) at load "P" (feet)	Breaking strength (minimum) (pounds)
1/2	1/16	7	185.0	530
3/4	1/16	12	83.5	1,050
1	1/16	20	48.0	1,700
1-1/4	1/16	38	30.0	2,500
1-1/2	1/16	50	21.0	3,350

3.2.2 Yarns. The plied yarns shall be either two-ply or three-ply ZS balanced-twist construction conforming to the requirements of table II.

TABLE II. Type II, yarns of balanced ply construction.

Designation		Feet per pound (min.)	Breaking strength (min.)
Type	Number of plies	Feet	Pounds
Marline	2	530	150
Marline	2	260	500
Houseline	3	470	440
Houseline	3	365	420
Roundline	3	165	750

3.3 Nonfibrous material. Extraneous material shall not be added for the purpose of weighing the ropes or plied yarns except coloring matter which is permissible in sufficient quantity to meet the requirements of 3.5. The moisture content of the ropes or plied yarns shall not exceed 2.0 percent. Finishing materials shall be permitted up to 30 percent of the dry weight of the completed product. The finishing materials shall be resistant to heat aging and shall not show strength losses greater than 10 percent (see 4.2.5.3).

3.4 Spliceability. The type I ropes shall be spliceable and shall not develop yarn displacement or strand cockles (see 4.2.5.5).

3.5 Color. Unless otherwise specified (see 6.2), the color of the finished ropes or plied yarns shall match an approved standard shade of OD 7.

3.5.1 Matching. The shade of the finished ropes or plied yarns shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7,500 degrees Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2,800 degrees Kelvin.

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3.5.2 Colorfastness. The finished ropes and plied yarns shall show fastness to weathering equal to or better than the standard sample after 40 hours exposure in accordance with method 5671 of FED-STD-191.

NOTE: Attention is called to contractors whose equipment operates at one cycle per 2 hours. This equipment may continue to be used. However, one cycle per minute shall be standard procedure in the event of dispute.

3.6 Identification ticket. In addition to the requirements specified above, each package unit of either type I or type II shall have a ticket attached with string or wire. The tickets shall be made of paperboard not less than 0.020 inch in thickness and the color shall be manila or light in intensity to permit the easy reading of the printed markings. The ticket shall be legibly printed with the following information:

(a) Stock number (if available).
(b) Nomenclature (including type and class).
(c) Specification number.
(d) Length.
(e) Contract number and date.
(f) Contractor's name.

3.7 Put-up.

3.7.1 Type I. Unless otherwise specified (see 6.2), ratline and seizing shall be furnished on nonreturnable reels in accordance with table III. Broken length will be permitted, but no piece shall be less than 600 feet when measured in the relaxed condition. The ends shall be cut off squarely and be securely whipped, taped, or heat sealed to prevent fraying or untwisting. The reels shall be so wound that each turn and layer is free from entanglement.

TABLE III. Net weight and length on reels for type I ropes.

Circumference (inches)	Minimum length (feet)	Approximate weight (pounds)
1/2	2,250	15
3/4	2,250	28
1	2,250	48
1-1/4	1,200	40
1-1/2	1,200	60

3.7.2 Type II. Unless otherwise specified (see 6.2), classes 1 and 2, type II plied yarns shall be packaged in 20-pound coils or tubes. The ends shall be cut off squarely and be securely whipped, taped or heat sealed to prevent fraying or untwisting.

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3.8 Workmanship. The ropes and plied yarns shall be of balanced construction evenly laid in accordance with high grade practice. The individual strands and the finished ropes shall be free from kinks and strand knots. There shall be no broken or loose ends projecting from the surface. There shall be no other manufacturing defects which will affect serviceability of the end product. The occurrence of defects shall not exceed the applicable acceptable quality levels specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Sampling and inspection shall be performed in accordance with MIL-STD-105, except where otherwise specified.

4.2.1 Inspection of components. Quality assurance provisions for specified components shall be in accordance with the applicable specifications and drawings to the extent applicable.

4.2.2 Examination of the end item for visual defects. The defects listed in table IV shall be enumerated regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample unit for this examination shall be one reel (spool) or coil, whichever is applicable. Ten percent of the gross length contained on each sample unit but not less than 100 feet shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels (spools) or coils each. The acceptable quality level shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

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TABLE IV. Visual examination defects.

Examine	Defect	Classification	
		Major	Minor
Appearance and workmanship	Cut, any (type I and II items)	x	
	Chafed or damaged, affecting serviceability (types I and II)	x	
	Kinks, broken or loose ends (type II only)	x	
	Kinks, darting yarns, 1/ broken or loose ends, bulged strands, strand knots (type I only)	x	
	Other than 3 strands (type I only)	x	
	Other than 2 ply yarn (type II, class 1)	x	
	Other than 3 ply yarn (type II, class 2)	x	
	Ends not cut off squarely (types I and II)		x
	Ends not securely whipped or taped (types I and II)		x
Identification ticket	Omitted, incorrect, illegible, insecurely attached		x
	Not as specified		x
Color	Not as specified	x	

1/ Darting yarns are internal yarns which project through the cover yarns of the strand at intervals along the rope (type I only).

4.2.3 Examination for length and winding. The sample unit shall be one spool, reel or coil. The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number 1. The lot size shall be the number of units in the inspection. Defects shall be as specified in 4.2.3.1 and 4.2.3.2.

4.2.3.1 Defects with regard to length. Defects with regard to length shall be considered to exist in either type I or II material if any of the following are found during inspection:

(a) Length of unit more than or less than specified length.
(b) Length of unit less than marked on ticket.
(c) Spools, coils or reels not in one continuous length.
(d) Any length on reels not less than 600 feet (type I only).

4.2.3.2 Defects with regard to winding. Defects with regard to winding shall be considered to exist if any of the following are found during inspection:

(a) Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of the rope or yarn.

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4.2.4 Examination of packaging requirements. An examination shall be made to determine conformance to preservation, packing and marking requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged but unsealed. Closure defects listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case-liners, container flaps, loose strapping or inadequate stapling. Bulging or distortion of containers.
Preservation	Not as specified. Spools or reels not secured, as specified.
Packing	Not as specified.
Weight	Gross weight in excess of requirements.

4.2.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table VI, shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The sample size, together with the acceptance and rejection numbers shall be in accordance with table V. The unit of product for testing shall be 50 feet of rope or yarn. The lot size shall be expressed in units of spools or reels.

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TABLE V. Sampling for tests.

Number of spools or reels in lot	No. of samples	Acceptance no. for each characteristic	Rejection no. for each test characteristic
15 and under	2	0	1
16 to 40	3	0	1
41 to 110	5	0	1
111 to 300	7	0	1
301 to 500	10	0	1
501 and over	15	1	2

TABLE VI. Test method. 1/

Characteristic	Requirement	Test method	No. of determinations per individual sample unit	Results reported as
Material	3.1	2/	1	Pass or fail
Strand of rope	3.2	Visual	1	Pass or fail
Ply of yarn	3.2	Visual	1	Pass or fail
Yarn denier	3.1	2/	1	Pass or fail
Direction of twist				
(a) Plied yarn	3.2	4050 of FED-STD-191	1	Pass or fail
(b) Rope	3.2	4050 of FED-STD-191	1	Pass or fail
Circumference				
(1) Type I only	3.2	4.2.5.1	3	Average of 3 determinations to nearest 1/16 inch.
Length per pound				
(a) Type I	3.2	4.2.5.2.1	1	Reported to nearest 0.01 foot
(b) Type II	3.2	4.2.5.2.2	1	Reported to nearest 0.01 foot
Breaking strength (initial)	3.2			
(a) Less than 600 pounds		4.2.5.3	3	Average of 3 determinations to nearest 10 pounds.
(b) More than 600 pounds		4.2.5.3	3	Average of 3 determinations to nearest 10 pounds.

See footnotes at end of table.

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TABLE VI. Test method. 1/ - Continued

Characteristic	Requirement	Test method	No. of determinations per individual sample unit	Results reported as
Breaking strength (after heat aging)	3.3	4.2.5.3.3	3	Average of 3 determinations to nearest 10 pounds.
Extractable matter	3.3	4.2.5.4	2	Average of 2 determinations to nearest 0.1 percent
Spliceability (type I only)	3.4	4.2.5.5	1	Pass or fail
Colorfastness to weathering	3.5.2	5671 of FED-STD-191	1	Pass or fail

1/ Tests to determine conformance to specification requirements including quantity of delivery may be made under prevailing atmospheric conditions except in settlement of dispute in which case the tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

2/ Certificates showing conformance to this specification shall be available to the inspector for each lot of rope or plied yarn. Certification shall contain actual test examination or other verifiable quality data.

4.2.5.1 Determination of circumference. The circumference for type I shall be measured at the beginning of the breaking strength test with the specimen under the load "P" specified in table I. A hard fiber shall be passed snugly around the rope and cut where it overlaps. The cut length shall be straightened and measured to the nearest 1/16 inch. This determination shall be repeated at least three times in different positions not less than two turns apart. The average of these determinations shall be the circumference.

4.2.5.2 Determination of length per pound.

4.2.5.2.1 Type I. When the circumference is specified, a minimum length of 20 feet shall be measured to the nearest 1/4 inch and weighed to within plus or minus 0.5 percent of its total weight. The feet per pound shall be calculated using the following equation:

$$\text{Feet per pound} = \frac{L \times \left(1 + \frac{e}{100}\right)}{W}$$

Where: L = Measured length of specimen (feet)
W = Measured weight of specimen (pounds)
e = Percent elongation at load P (see table I, determined as specified in 4.2.5.3.2)

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4.2.5.2.2 Type II. Where the circumference is not specified, not less than 100 feet shall be accurately measured with just sufficient tension to keep the specimen barely taut. The specimen shall then be weighed and the feet per pound calculated using the following equation:

$$\text{Feet per pound} = \frac{L}{W}$$

Where: L = Measured length of specimen (feet)
W = Measured weight of specimen (pounds)

4.2.5.3 Determination of breaking strength.

4.2.5.3.1 Less than 600 pounds. When the specified minimum breaking strength is less than 600 pounds, the test shall be conducted in accordance with method 6016 of FED-STD-191.

4.2.5.3.2 Greater than 600 pounds. When the specified minimum breaking strength is greater than 600 pounds, the test shall be conducted in accordance with method 6015 of FED-STD-191 except that load "P" shall conform to the requirements of table I.

4.2.5.3.3 Breaking strength after heat aging. Three breaking strength specimens of each type and item shall be heated for 5 days in a convection air oven at 175 ± 2 degrees Fahrenheit (°F). The specimens shall be allowed to reach equilibrium at standard conditions before being tested for breaking strength. The percent change in breaking strength shall be calculated based on the average initial strengths and after heat aging.

4.2.5.4 Extractable matter. A sample rope or plied yarn weighing approximately 10 grams shall be placed in a tared weighing bottle and dried to constant weight at 105 to 110 degrees Celsius (°C). After the bone dry weight has been determined, the sample shall be transferred to a Soxhlet extraction apparatus for a minimum of 20 extractions using diethyl ether or other solvent. The ether shall then be evaporated from the extract in the tared flask at a temperature of 105 ± 5°C to constant weight, cooled in a desiccator and weighed. This is the weight of extractable matter. The percent extractable matter present shall be calculated by use of the following formula:

$$\text{Percent extractable matter} = \frac{\text{Weight of extractable matter}}{\text{Weight of dry specimen}} \times 100$$

4.2.5.5 Determination of spliceability. A breaking strength specimen of type I rope shall be prepared and spliced at each end with at least four full tucks. The specimen shall then be allowed to rest 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the type I rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

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5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. The rope and plied yarns shall be preserved in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, or C. The rope and plied yarns shall be packed in accordance with the applicable requirements of MIL-C-3131. When specified (see 6.2) reels or coils of rope or plied yarns shall be palletized in accordance with MIL-STD-147.

5.3 Marking. In addition to any special markings required (see 6.2), interior containers and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rope and yarn covered by this specification is intended for use as ratline, seizing, marline, houseline, and roundline when acquisition of natural fibers such as hemp and jute is not possible.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Color (see 3.5).
(d) Put-up and length when other than specified (see 3.7.1 and 3.7.2).
(e) Levels of preservation and packing (see 5.1 and 5.2).
(f) Whether packaged rope shall be palletized (see 5.2.1).
(g) Special marking required (see 5.3).
(h) That the rope shall be purchased on a price-per-pound basis net weight.
(i) That the rope plied yarns shall be purchased on a price-per-pound basis net weight.
(j) That contracting activity will accept, at original weight, any unit which has been shortened or cut for test specimens, if in accordance with this specification.

6.3 Standard sample. For access to the standard sample, address the acquisition office issuing the invitation for bids.

6.4 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

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6.5 Subject term (key word) listing.

Colorfastness
Houseline
Marline
Monofilament
Multifilament
Ratline
Seizing

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-24183A(SH)	ROPE AND YARN, PLIED, SYNTHETIC FIBER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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